

Problem 1 (NIMO Winter 2014/3)

We claim the maximum is $\sqrt{384}$. This can be achieved as follows:

$$8, 9, 10 \rightarrow \sqrt{245}$$

$$6, 7, \sqrt{245} \rightarrow \sqrt{330}$$

$$4, 5, \sqrt{330} \rightarrow \sqrt{371}$$

$$2, 3, \sqrt{371} \rightarrow \sqrt{384}$$

We remain with 1, $\sqrt{384}$ and can merge no more.

To prove that this is the maximum, note that the sum of the squares of the numbers is invariant. During an operation, the sum of the squares of the numbers being merged and the square of the result is the same.

$$a^2 + b^2 + c^2 = \left(\sqrt{a^2 + b^2 + c^2} \right)^2$$

At the end, we must have 2 numbers remaining on the board since we reduce the number of numbers on the board 2 at a time. To maximize one of these numbers, the other number must be $\sqrt{1}$, the lowest possible number. The other number is hence $\sqrt{\sum_{i=1}^{10} i^2 - 1} = \sqrt{384}$.

Problem 2 (Shortlist 2018 N1)

We claim that all pairs (m, n) where $m \nmid n$ and $n \nmid m$, or $m = n$, and are the only pairs that work.

We start with proving that no other pairs work. For $m \mid n$ or $n \mid m$ but $m \neq n$, we WLOG $m \mid n$. The factors of sn includes all the factors of sm but also additionally has n itself. Therefore, sn always has strictly more factors than sm for any positive s so it is impossible such pairs to work.

We now prove that all such pairs work. The $m = n$ case is trivial, so if $m \nmid n$ and $n \nmid m$, we write,

$$m = 2^{j_1} 3^{j_2} 5^{j_3} \dots; n = 2^{k_1} 3^{k_2} 5^{k_3} \dots; s = 2^{m_1} 3^{m_2} 5^{m_3} \dots$$

We let $k = \{k_1, k_2, k_3, \dots\}$ and $j = \{j_1, j_2, j_3, \dots\}$. The number of factors sm has is $\prod (j_i + m_i + 1)$ and the number of factors sn has is $\prod (k_i + m_i + 1)$. We are proving these can be made equivalent with a certain choice of $m_1, m_2, \dots$

$$\begin{aligned} \prod (j_i + m_i + 1) &= \prod (k_i + m_i + 1) \\ \Leftrightarrow \prod \frac{j_i + m_i + 1}{k_i + m_i + 1} &= 1 \Leftrightarrow \prod \frac{k_i + m_i + 1 + (j_i - k_i)}{k_i + m_i + 1} = 1 \end{aligned}$$

We let $r_i := \frac{k_i + m_i + 1}{j_i - k_i}$.

$$\Leftrightarrow \prod \frac{r_i + 1}{r_i} = 1$$

Whenever $j_i > k_i$, r_i is positive. Whenever $j_i < k_i$, r_i is negative. Since $m \nmid n$ and $n \nmid m$, we have at least one case of each. We let P be the number of positive r_i , N be the number of negatives, and $x = \max(k_i + 1)$.

We let $a = \{a_0, a_1, \dots, a_{P-1}\}$ be the positive r_i s and $b = \{b_0, b_1, \dots, b_{N-1}\}$ be the negative r_i s.

$$\Leftrightarrow \prod \frac{a_j + 1}{a_j} \cdot \prod \frac{b_k + 1}{b_k} = 1 \quad (a)$$

Now, there is always a way to change m_i such that $a_j = xP + j$. Therefore,

$$\prod \frac{a_j + 1}{a_j} = \frac{xP + P}{xP}$$

There is also a way to change m_i such that $b_j = -xN - j$. Therefore,

$$\prod \frac{b_j + 1}{b_j} = \frac{-xN}{-xN - N}$$

Therefore, (a) is satisfied as,

$$\frac{-xN}{-xN - N} \cdot \frac{xP + P}{xP} = \frac{x^2NP + xNP}{x^2NP + xNP} = 1$$

Problem 3 (USAMO 2011/2)

We first prove that there is no more than one winning vertex. We can let a_i be the value of the i -th vertex. Let,

$$g = a_1 + 2a_2 + 3a_3 + 4a_4$$

Note that the value of $g \bmod 5$ never changes over the operations. Therefore, depending on the initial value of $g \bmod 5$, we can determine what cannot be the winning vertices. For example, if $g \equiv 2 \bmod 5$, then a_0, a_1, a_3, a_4 cannot be winning since, in the end, these would respectively result in being $0, 1, 3, 4 \bmod 5$.

We now prove by construction there is always at least one winning vertex.

WLOG, if the first vertex is determined to be the only one that can be won, we can do the following.

We let $x_i := f_i - a_i$ where a_i is the initial number at vertex i , f_i is the final value, and so x_i is the cumulative gain of vertex i . Therefore, in the case where we win at vertex 0,

$$2011 - a_0 = 2x_0 - x_2 - x_3;$$

$$-a_1 = 2x_1 - x_3 - x_4;$$

$$-a_2 = 2x_2 - x_4 - x_0;$$

$$-a_3 = 2x_3 - x_0 - x_1;$$

$$-a_4 = 2x_4 - x_1 - x_2$$

Notice that if $(x_0, x_1, \dots, x_4)$ is a solution, then $(x_0 + k, x_1 + k, \dots, x_4 + k)$ is also a solution. This makes sense since if we take a turn with the same integer m on each of the vertices, we just get back to the beginning. Therefore, WLOG, let $x_0 = 0$.

If we solve the equations, we get,

$$x_0 = 0;$$

$$x_1 = \frac{4a_0 - 2a_1 + 2a_2 + a_3 - 8044}{5};$$

$$x_2 = \frac{3a_0 + a_1 - a_2 + 2a_3 - 6033}{5};$$

$$x_3 = \frac{2a_0 - a_1 + a_2 - 2a_3 - 4022}{5};$$

$$x_4 = \frac{3a_2 + 6a_0 + 2a_1 + 4a_3 - 12066}{5};$$

For the first vertex to be the winning vertex, we must have gotten,

$$a_1 + 2a_2 + 3a_3 + 4a_4 \equiv 0 \pmod{5}$$

We also know that $a_0 + a_1 + a_2 + a_3 + a_4 = 2011 \equiv 1 \pmod{5}$.

Therefore,

$$4a_0 - 2a_1 + 2a_2 + a_3 - 8044 \equiv 4(a_0 + a_1 + a_2 + a_3 + a_4) + 4(a_1 + 2a_2 + 3a_3 + 4a_4) + 1 \equiv 0 \pmod{5};$$

$$3a_0 + a_1 - a_2 + 2a_3 - 6033 \equiv 3(a_0 + a_1 + a_2 + a_3 + a_4) + 3(a_1 + 2a_2 + 3a_3 + 4a_4) + 2 \equiv 0 \pmod{5};$$

... rest omitted for brevity (would write out if this were actual contest).

Hence, each of $x_0, \dots, x_4$ are integers, we are done.

Problem 5 (JMO 2016/4)

We claim that the minimum is 6097392.

Proof 6097392 works:

$$6097392 = 1008 \cdot 6049$$

We can write 6049 in 3024 ways as follows:

$$(1 + 6048), (2 + 6047) \dots, (3024 + 3025)$$

Only up to 2016 of these pairs will have at least one of its two numbers removed, leaving 1008 pairs of 2 numbers that each sum to 6049. We are hence done.

Proof that anything below 6097392 doesn't work:

For any $N < 6097392$, we take away the numbers $1, \dots, 2016$. The minimum sum of 2016 distinct numbers the remaining numbers can have is 6097392, meaning we cannot construct N .

Problem 6 (USEMO 2021/1)

We call cells that are strictly less than the average in its row and greater than the average in its column (the cells we want to count) *rainbows*.

We claim a maximum of $(n - 1)^2$ *rainbows*.

Proof that $(n - 1)^2$ is attainable:

1	1	...	1	2
1	1		1	2
	$\vdots$	$\ddots$	1	2
1	1	1	1	2
0	0	0	0	69.420

In this $n \times n$ grid, each of the 1's in the $n - 1 \times n - 1$ square on the top-left are *rainbows*.

Proof that $(n - 1)^2$ is the upper bound:

For any cell in the grid with value x , we call x a *failure* if it is the number closest to the bottom of the grid that ties for the least in its column and a *nerd* if it is the number closest to the right of the grid that ties for the most in its row. If x is either of these, then x cannot be a *rainbow*.

We claim there to be at least $2n - 1$ cells which are either *failures* or *nerds*, meaning there can be no more than $(n - 1)^2$ *rainbows*.

Since there is exactly one *failure* in each column and one *nerd* in each row, there are $2n$ *failures* and *nerds* combined. Therefore, we must prove there can be no more than 1 cell that is both a *failure* and a *nerd*, and we'll be done with PIE.

For the sake of contradiction, let there be two distinct *failure-nerds* in the grid. Obviously, these two cannot be in the same row or column. WLOG, let them be respectively the top-left and bottom-right cells.

We let the values of the top-left, top-right, bottom-left, and bottom-right numbers be a, b, c, d respectively. We have $a < c \leq d; a > b \geq d; \Rightarrow a < d; a > d$, a contradiction.

23 clubs up to this point; 5 hours cumulative (including examples)

Problem 10 (Slovenia TST 2018)

We prove a stronger conclusion by induction on c : Given cn marbles of at most c colors, we prove they may be placed in c boxes with n marbles in each box such that each box contains marbles of at most two colors.

Base Case: $c = 1$. This is trivial, all marbles in one box of at most one color.

Inductive Step: From $c = k$ to $c = k + 1$.

Let $x_1, x_2, \dots, x_{k+1}$ be the counts of the marbles of each color. By Pigeonhole, since the average of these numbers is n ,

$$x_i = \min(x_1, x_2, \dots, x_{k+1}) \leq n \leq \max(x_1, x_2, \dots, x_{k+1}) = x_j$$

We put all x_i of the least occurring marbles with $n - x_i$ of the most-occurring marbles. Since,

$$0 \leq n - x_i \leq n \leq x_j$$

This is a valid move. We are then left with nk marbles of k colors, so we're done.

Conclusion: The claim holds for any c, n .

Problem 11 (Kazakhstan 2016, added by Tilek Askerbekov)

SKETCH:

We write n as $n = p_0^{e_0} p_1^{e_1} \dots p_x^{e_x}$. This is equivalent to having a non-self-intersecting path in a x -dimension hypercube of dots of dimension $e_0 + 1 \times e_1 + 1 \times \dots \times e_x + 1$ where one only moves one step to an adjacent non-diagonally dot.

We let these dots be represented as $(c_1, \dots, c_n) \in \{(1, 1, \dots, 1), \dots, (e_0 + 1, \dots, e_x + 1)\}$.

We prove an even stronger statement using induction on x , that you can start at any corner and end up at another corner.

Base Case: $x = 1$, trivial, move in the line to the only other corner.

Inductive Step: If it's proven for $x = k$, we can isolate c_1 , and snake through all the dots in the grid where $c_1 = 1$ and end up at another corner. We then increment c_1 until we have hit every point.

Conclusion: This holds for all $x \geq 1$. Since all number ≥ 2 have at least one prime factor we're done.

Problem 13 (PAGMO 2021/5)

We claim that this is possible for any configuration if and only if $3 \nmid n$.

Proof that if $3 \nmid n$, it is possible:

We can do the following construction to take any number from $k + 3$ to k (modulo n):

$$\{k+3\} \rightarrow \{k+2, k+4\} \rightarrow \{k+1, k+3, k+4\} \rightarrow \{k, k+2, k+3, k+4\} \\ \rightarrow \{k, k+2, k+2, k+4, k+4\} \rightarrow \{k\}$$

Therefore, if $3 \nmid n$, we can perform these steps and take any number to 1. If $2 \nmid m$, we first eat any candy k and replace it with $k-1, k+1$ to have an even m . Then, we do this to every candy on the board right now until we have an even number of 1's, at which time we eat them all pair by pair.

Proof that if $3 \mid n$, it is impossible:

We call a candy of type k where $k \equiv 0, 2 \pmod{3}$ an *ilost*, and candies of type $k \equiv 1 \pmod{3}$ the *game*.

On each move, the number of *ilost*'s changes by an even number. Therefore, if we start with exactly one *ilost*, we cannot finish.

38 clubs up to this point; 7.5 hours cumulative (including examples)

Problem 16 (Canada 2019/3)

WLOG, let the bottom-left and top-right corner be white squares. We partition the chessboard into mn 2×2 squares that are $\begin{pmatrix} \text{black} & \text{white} \\ \text{white} & \text{black} \end{pmatrix}$.

We can only place counters on one of the white squares in each 2×2 . Since there are only mn of these 2×2 in total, **we must place exactly one counter on each to place mn counters.**

We can express our placement as a binary $m \times n$ grid g of 0's and 1's. A 0 means that, in the corresponding 2×2 square in the original grid, the counter is placed at the bottom-left square. A 1 means that the counter is placed at the top-right.

We have a bijection from g to our original grid, so we just count the number of valid g s.

Obviously, each row of g must be increasing from left-to-right, and each column must be increasing from bottom up.

We claim that the total is $\binom{m+n}{m}$ by induction.

Base Case: When $m = 0$ or $n = 0$, then we have one way as desired.

Inductive Step: Count (m, n) using $(m-1, n)$ and $(m, n-1)$.

If our first row is all 1, then we have $(m-1, n)$. Otherwise, we have $(m, n-1)$. Therefore, our total count is, by Pascal's,

$$f(m, n) = f(m-1, n) + f(m, n-1) = \binom{m+n-1}{m-1} + \binom{m+n-1}{n-1} = \binom{m+n}{m}$$

Conclusion: We are done by mathematical induction.

Problem 19 (JMO 2021/4)

Claim 1: After x moves, the maximum area of ABC is $\frac{\frac{(x+1)}{2} \cdot \frac{(x-1)}{2}}{2}$ if x is odd, otherwise $\frac{\frac{x}{2} \cdot \frac{x}{2}}{2}$.

Construction: If x is odd, move A in the $+x$ direction of the x -axis $\frac{(x+1)}{2}$ times and B in the $+y$ direction $\frac{x-1}{2}$ times. If x is even, do the same but $\frac{x}{2}$ times each instead.

Proof of maximum: Define the bounding box of ABC to be the smallest axis-parallel rectangle which contains all three vertices A, B, C . On each move, the width of ABC can increase by up to 1, and the length of ABC can increase by up to 1, but not both at the same time. Therefore, by AM-GM, after x moves, the area of this bounding box can be up to $\frac{(x+1)}{2} \cdot \frac{(x-1)}{2}$ if x is odd and $\left(\frac{x}{2}\right)\left(\frac{x}{2}\right)$ if x is even.

Note that since ABC is a triangle inscribed in a rectangular bounding box, its area is no more than half of the bounding box. We are hence done.

Claim 2: The answer is 128.

Proof of minimum: Using *Claim 1*, we can get that $x \geq 128$ for the area to be at least 2021.

Construction: Move A to $(-53, -1)$, B to $(0, 64)$ and C to $(10, 0)$.

Problem 21 (USACO 2008/4)

We claim that the answer is all numbers n of the form $n = 2^i + 2^j$ for $i, j \geq 0$.

Construction: If $i, j > 0$, draw a line from one vertex to another vertex that is 2^i sides away from it. Let the vertices on either side of this line be named.

$$a_0 = b_0, a_1, b_1, a_2, b_2, \dots, a_{2^i-1}, b_{2^j-1}, a_{2^i} = b_{2^j}$$

We first draw the lines $a_0a_2, a_2a_4, \dots, a_{2^i-2}a_{2^i}, b_0b_2, b_2b_4, \dots, b_{2^j-2}b_{2^j}$. We then do the same for $a_0a_4, a_4a_8, \dots, a_{2^i-4}a_{2^i}, b_0b_4, \dots, b_{2^j-4}b_{2^j}$. We repeat this for both a and b until we have re-drawn $a_0a_{2^i} = b_0b_{2^j} = \text{initial line}$. Every one of the triangles we form are equilateral.

Proof all other n are impossible: **(SKETCH)**

If we have a polygon with k equal sides and $k - 1$ equal angles, with one other side of another length (which we call quirky). We claim that we can only divide it into isosceles triangles if $k = 2^i$ for some positive i .

Since every side of k must be part of an isosceles triangle, the quirky side be too. The other vertex must be the one that the perpendicular bisector of the quirky side passes through. Therefore, $2 \mid k$. Repeat to get result.

For $n \geq 4$, we must create at least one diagonal. No matter what the diagonal is, if $n \neq 2^i + 2^j$, we can't do it.